

## □ CHAPTER 2: Requirement Analysis Specification

### 2.1 Functional System Requirements

- Applicant Context Ingestion: The runtime core must consume multidimensional profile arrays safely mapping to structural fields including gender, marital status, education level, dependency indexes, and property area coordinates.
- Financial Transaction Tracking: The engine must process high-variance numeric attributes including primary applicant base income, coapplicant revenue parameters, exact requested loan volumes, and temporal repayment cycles.
- Algorithmic Selection Matrix: The controller layer must allow real-time selection switching across a pipeline composed of Decision Tree, Random Forest, KNN, and XGBoost.
- Analytical Ingestion Sink: The backend must output clear prediction results consisting of a binary status state (Approved / Rejected) backed by an exact continuous probability\_score variable.

### 2.2 Non-Functional System Capabilities

- Environmental Portability: The core code boundary must remain perfectly host-agnostic, running reliably inside isolated, lightweight Docker containers.
- Sub-Second Performance Limits: Computational inference evaluation times must execute in under 100 milliseconds under high concurrent transaction simulation.
- Responsive Layout Standards: The client presentation interface must adapt seamlessly to varying viewport sizes across mobile devices and desktop workstations.